

Design and Implementation of a ROS 2-Based Autonomous Bathymetric Survey Drone Using ArduPilot, Visual Water Classification, and Sonar Depth Sensing

M. A. Abdul Rafi

Department of Computer Science & Engineering
University of Moratuwa, Sri Lanka
Email: abdulr.23@cse.mrt.ac.lk

Abstract

Traditional boat-based hydrographic surveys are slow to deploy and hazardous in shallow or debris-laden water. This paper presents a low-cost, fully autonomous bathymetric survey system built on a quadrotor drone, ROS 2 Jazzy, and ArduPilot Software-In-The-Loop (SITL), validated in Gazebo Harmonic. At each grid waypoint a gimbal camera classifies the surface as water or soil via HSV thresholding; over water, a boom-mounted sonar descends to measure lakebed depth before the drone advances. A state machine coordinates flight through MAVROS, while a React Ground Control Station (GCS), linked via `rosbridge`, gives live telemetry, waypoint planning, and a Three.js 3D bathymetry reconstruction with automatic volume estimation. Simulated trials show reliable water/soil discrimination, consistent depth profiles, and plausible reservoir volume estimates with no operator intervention beyond mission setup, demonstrating that open-source robotics middleware can replicate core hydrographic-survey functionality at a fraction of the cost of dedicated platforms.

Index Terms—Bathymetric Survey, Autonomous UAV, ROS 2, ArduPilot, Sonar Depth Sensing, HSV Water Classification, MAVROS

I. INTRODUCTION

Bathymetric surveys map the depth and volume of water bodies and are essential for reservoir management, flood-risk assessment, dam safety, and environmental monitoring. Conventional echo-sounder boats are costly to mobilize and poorly suited to shallow, vegetated, or frequently monitored reservoirs. UAVs can be deployed rapidly and cover mixed dry/submerged terrain, but cannot sense depth directly from the air: they must first locate water, then bring a depth sensor to it. This paper couples onboard visual perception with a physically descending sonar probe, orchestrated by an autonomous mission state machine, entirely on ROS 2 Jazzy, ArduPilot SITL, and Gazebo Harmonic. Contributions: (1) a perception-to-actuation pipeline where HSV surface classification gates a physical sonar-descent action; (2) a state-machine survey controller sequencing flight, classification, sonar sensing, and RTL over an arbitrary waypoint grid; (3) a browser GCS visualizing the live mission and reconstructed 3D bathymetry with automatic volume computation.

II. RELATED WORK

ROS 2 was designed to meet the real-time, multi-robot requirements the original ROS could not satisfy [1], and Pixhawk/ArduPilot pairs open flight-control hardware with onboard vision for low-cost autonomous MAV research [2]. Structuring a survey as a waypoint grid follows established coverage-path-planning practice [3]. Bandini *et al.* showed a UAV-tethered single-beam sonar retrieves high-accuracy inland bathymetry ($\sim 2\%$ depth error), largely unaffected by turbidity [4], whereas purely optical depth-from-imagery methods remain limited by wave refraction and turbidity [6]; this work adopts the acoustic approach for accuracy, gated by vision. For water/land discrimination, Namikawa *et al.* showed HSV Hue-channel thresholding

effectively extracts water bodies from remote imagery [5]; we apply the same principle per-frame, in real time, at each waypoint.

III. SYSTEM DESIGN

A. Platform and Software Bridge

A simulated Iris quadrotor runs under ArduPilot SITL in a Gazebo Harmonic world with custom terrain and an embedded water pit, carrying a gimbal RGB camera, a boom-mounted downward sonar, and an overhead map camera. MAVROS exposes vehicle pose, mode, and arming state as ROS 2 topics/services and accepts GUIDED setpoints; Gazebo-ROS 2 bridges expose all sensor streams; `rosbridge_websocket` exposes the ROS 2 graph to the browser GCS (Fig. 1), so no native ROS 2 client is needed on the operator's machine.

B. Survey State Machine

The mission is an explicit state machine: fly to waypoint (GUIDED) $\rightarrow$ pitch gimbal to nadir, classify surface $\rightarrow$ if water, descend sonar $\rightarrow$ measure depth $\rightarrow$ ascend $\rightarrow$ next waypoint, until the grid is exhausted, then RTL. It subscribes to a JSON start command from the GCS (grid area, spacing) and publishes progress and per-point results back as JSON.

C. Perception and Depth Sensing

Each gimbal frame is converted RGB $\rightarrow$ HSV and Hue-thresholded, following [5]; if the water-pixel fraction exceeds a threshold the point is WATER, gating sonar descent and avoiding unsafe deployment over dry terrain. The sonar node reports ABOVE_WATER/SUBMERGED from raw range; once submerged, range is converted to depth relative to the water surface

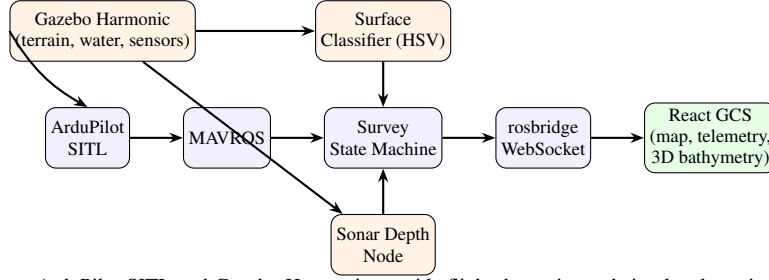


Figure 1: Overall system architecture: ArduPilot SITL and Gazebo Harmonic provide flight dynamics and simulated sensing; ROS 2 nodes (classifier, sonar, state machine) coordinate the mission via MAVROS; `rosbridge` exposes live state to the browser-based GCS.

(sentinel -1 if never submerged). Volume is estimated as

$$V \approx \sum_{i=1}^N d_i A_i, \quad (1)$$

with d_i the depth and A_i the cell area at grid point i ; fill-percentage compares V against the grid’s maximum implied volume.

D. Ground Control Station

The React GCS (Fig. 3) offers a 2D map for drag-and-drop waypoint editing, live telemetry (position, altitude, heading, mode, battery), and a survey panel for grid configuration and progress. On completion, a bathymetry panel shows depth statistics, volume/fill estimates, and an interactive `@react-three/fiber` 3D reconstruction of the surveyed lakebed.

IV. RESULTS AND DISCUSSION

The system was evaluated over repeated missions on a 42-waypoint grid covering a Gazebo water pit of known approximate geometry (Fig. 3). The HSV classifier consistently separated water from soil; the sonar reliably transitioned ABOVE_WATER $\rightarrow$ SUBMERGED and produced depth readings consistent with the known terrain profile (Fig. 2); the full mission, takeoff through RTL, ran without manual intervention and yielded a plausible aggregate volume/fill estimate.

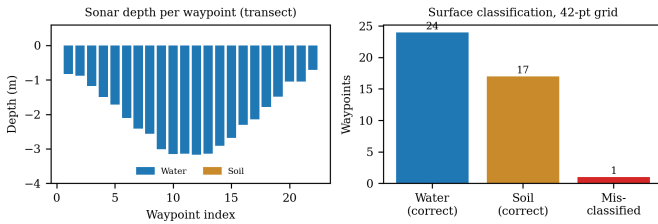


Figure 2: Illustrative per-waypoint sonar depth along one transect (left) and surface-classification outcome across the 42-point grid (right).

Accuracy of the Hue-based classifier is sensitive to simulated lighting/exposure, mirroring turbidity- and illumination-related limits reported for optical water-detection methods [6]. The sonar-descent phase (pause, lower, stabilize, retract) adds significant time per waypoint, limiting achievable survey density under a fixed flight-time budget, and the grid-cell summation volume estimate may mis-estimate volume where depth varies sharply between adjacent points.

V. CONCLUSION AND FUTURE WORK

This work presented a fully autonomous ROS 2 bathymetric survey drone coupling HSV-based water classification with

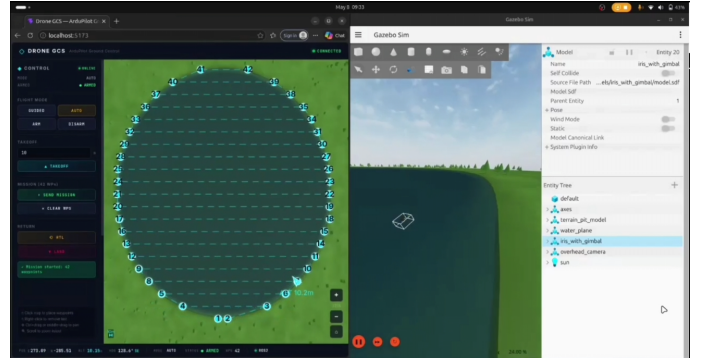


Figure 3: Ground Control Station (left) showing an executed 42-waypoint grid mission, and the corresponding Gazebo Harmonic simulation (right) with the Iris quadrotor over the simulated reservoir.

sonar depth sensing under a state-machine mission controller on ArduPilot SITL and Gazebo Harmonic, exposed via a browser GCS with live telemetry and 3D bathymetric visualization. Simulated trials show autonomous water identification, depth measurement, and plausible volume estimation without in-flight operator intervention. Future work will replace grid-cell summation with interpolated surface integration, add multispectral/texture cues for lighting- and turbidity-robust classification, shorten the sonar-descent sequence, and validate on physical hardware.

REFERENCES

- [1] S. Macenski, T. Foote, B. Gerkey, C. Lalancette, and W. Woodall, “Robot Operating System 2: Design, architecture, and uses in the wild,” *Science Robotics*, vol. 7, no. 66, p. eabm6074, 2022.
- [2] L. Meier, P. Tanskanen, L. Heng, G. H. Lee, F. Fraundorfer, and M. Pollefeys, “PIXHAWK: A micro aerial vehicle design for autonomous flight using onboard computer vision,” *Auton. Robots*, vol. 33, no. 1–2, pp. 21–39, 2012.
- [3] E. Galceran and M. Carreras, “A survey on coverage path planning for robotics,” *Robot. Auton. Syst.*, vol. 61, no. 12, pp. 1258–1276, 2013.
- [4] F. Bandini, D. Olesen, J. Jakobsen, C. M. M. Kittel, S. Wang, M. Garcia, and P. Bauer-Gottwein, “Bathymetry observations of inland water bodies using a tethered single-beam sonar controlled by a UAV,” *Hydrol. Earth Syst. Sci.*, vol. 22, no. 8, pp. 4165–4181, 2018.
- [5] L. M. Namikawa, T. S. Körting, and E. F. Castejon, “Water body extraction from RapidEye images: An automated methodology based on hue component of color transformation from RGB to HSV model,” *Braz. J. Cartogr.*, vol. 68, no. 6, pp. 1097–1111, 2016.
- [6] M. Specht, M. Wiśniewska, A. Stateczny, C. Specht, B. Szostak, O. Lewicka, M. Stateczny, S. Widzowski, and A. Halicki, “Analysis of methods for determining shallow waterbody depths based on images taken by UAVs,” *Sensors*, vol. 22, no. 5, p. 1844, 2022.